Description of Bladed model of IEA 15 MW turbine

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1 INTRODUCTION

The current document presents a description of a Bladed model of the IEA 15MW turbine *IEA-15-240-RWT* (version 1.0), which was originally defined in IEA Wind Task 37 for HAWC2 and OpenFAST.

Section 2 presents a high-level description and summarises the sources of data employed for building the Bladed model. Section 3 gives an overview of the control dynamics. Section 4 provides the areas where the model has been verified. Section 5 describes the files in the package supplied by DNV to run this model.

Note that the current Bladed model corresponds to the release version 1.0 from the IEA Task 37. An updated version (1.1) will be released at a later stage.

2 MODEL DESCRIPTION

DNV has built a Bladed model of the IEA 15MW reference turbine. The model is described in the IEA Task 37 report /1/ and additional information is available in the the project's GitHub site /2/.

Main turbine properties are summarised in Table 2-1 and section 2.1 provides an overview of the sources of data employed and any assumptions taken in the Bladed model built.

Table 2-1Turbine specifications

Parameters	IEA Wind 15-MW Turbine	Units
Power rating	15	MW
Specific rating	332	W/m^2
IEC Turbine Class	IB	-
Rotor orientation	Upwind, clockwise	-
Number of blades	3	-
Control type	Variable speed	-
Drivetrain	Direct Drive	-
Cut-in wind speed	3	m/s
Cut-out wind speed	25	m/s
Design TSR	9	-
Minimum rotor speed	5	rpm
Maximum rotor speed	7.56	rpm
Rotor diameter	240	m
Rotor precone angle	-4	
Rotor tilt angle	6	deg
Blade prebend	3.99	m
Hub height	150	m
Transition piece height	15	m
Water depth	30	m
Monopile embedment depth	45	m
Blade mass	65	t
RNA mass	1,015	t
Tower mass	860	t
Monopile mass	1,318	t

2.1 Summary of employed data sources

2.1.1 Rotor-nacelle assembly (RNA)

Main RNA properties were taken directly from the IEA Task 37 report /1/ and the *RNA_modeling_properties.xlsx* spreadsheet from the documentation folder in the project's GitHub site /3/. Exceptions are:

- Blade set angle - It has been assumed to be zero, since no value was reported.
- Over-hang – There was conflicting information between the value reported in the IEA report and all the spreadsheets within the documentation folder (*RNA_modeling_properties.xlsx, Nacelle_Mass_Properties.xlsx, IEA-15-240-RWT_FineGrid_tabular.xlsx*). The value most cited in the documentation was adopted in the Bladed model.
- Hub vertical offset – It has been estimated to match specified hub-height of 150m.
- Drivetrain torsional damping – It has been adjusted so the shaft has 5% of critical modal damping, as cited in the IEA report.

2.1.2 Blades

To build the blades, all aerodynamic and structural properties were taken from the HAWC2 model /4/, except for the shear centre and mass moment of inertia, which were taken from the OpenFast model /5/.

The IEA report /1/ mentions assumed blade structural damping of the first flapwise, edgewise and torsional modes of 3%, 3% and 6%, respectively. It has been confirmed with NREL that these are the ratios from critical modal damping and therefore included in the Bladed model.

2.1.3 Tower

The tower geometrical properties (height, diameter and wall thickness) were taken directly from the IEA report /1/ and the sections' structural properties were calculated with Bladed using the Young's and shear modulus material properties in the IEA report. The steel density was increased by 7% to account for outfitting mass beyond the cylindrical shell mass, as indicated in the IEA report.

It should be noted that discrepancies were observed between the OpenFAST model /5/, *IEA-15-240-RWT_FineGrid_tabular.xlsx* spreadsheet in documentation folder /3/ and those plotted in the IEA report /1/. Bladed calculated values were in good agreement with the ones in the IEA report /1/.

Foundation horizontal translational and rotational stiffnesses were included in the model using the soil properties in the IEA report /1/ and applying the equations from Appendix B of the report.

The IEA report /1/ mentions a tower structural damping of the first fore-aft, side-side and torsional modes of 2%.

2.1.4 Control

Control parameters defined in the IEA report /1/ were used, which are in line with those calculated with Bladed with rigid blades (see section 4.2). Note that the dynamic controller parameters have been re-tuned as described in section 3.

2.2 Missing or unusual model parameters

The turbine model as defined in IEA Task 37 covers many key turbine parameters. However, some properties that would typically be specified in a Bladed model for commercial use have not been specified.

Below is a summary of missing information that would usually be defined

- The **shaft brake** strength has not been defined. To run *Parked* cases, a brake must be defined. We have assumed a large value of 100MNm so that the rotor won't turn during Parked cases.
- No **pitch actuator** information has been provided, so a very basic actuator definition, with a passive transfer function response has been defined. Typically pitch actuator details such as gear ratio, torque limits and friction would be defined to give a more accurate pitch response. Setpoint trajectory planning (acceleration and velocity limits on change in pitch demand) remain undefined, which could lead to high pitch acceleration if the controller pitch demand changes quickly.
- **Blade modal damping** ratio is higher than typically used at 3% for flapwise and edgewise modes, and 6% on the torsional mode. Typical values used in commercial projects are around 0.5-1.1% structural damping ratio. Similar for **tower modal damping**.
- Discretisation of **monopile structural / hydrodynamic mesh** around waterline is 5m node spacing. Wave loads would be resolved more accurately if this was reduced to ~1-2m near the waterline.

2.3 Summary of main Bladed model settings

Table 2-2 presents a summary of the main Bladed model settings.

Table 2-2: Summary of Bladed settings

Bladed interface version	4.11.0.11
DTBLADED version	4.11.0.11
Wake model	Oye dynamic wake model
Skew wake correction model	Glauert skew wake model
Stall model	Dynamic stall according to Incompressible Beddoes-Leishman starting at 25% radius and ending at 95% radius
Modal model	Blade modelled as multi-part: 5 parts, with 10 modes per part, leading to a total of 50 blade modes 8 support structure modes
Tower shadow model	Potential flow model

3 CONTROL DYNAMICS

The ROSCO controller has been re-tuned to work with the Bladed model of the IEA 15MW turbine.

The reason this was required is that the Bladed model was configured to have a significant (realistic) lag in the pitch actuator response, which was not accounted for in the original tuning. The turbine model was linearised in Bladed and the linear model imported to Matlab to perform control tuning. The only control loop that was re-tuned was the pitch-speed loop. Now at 10m/s, that loop has a gain margin of 9.6dB and a phase margin of 64°. At 20m/s it has a gain margin of 12dB and a phase margin of 65°. This gives the pitch controller a fairly quick step response with very little overshoot as shown in Figure 3-1.

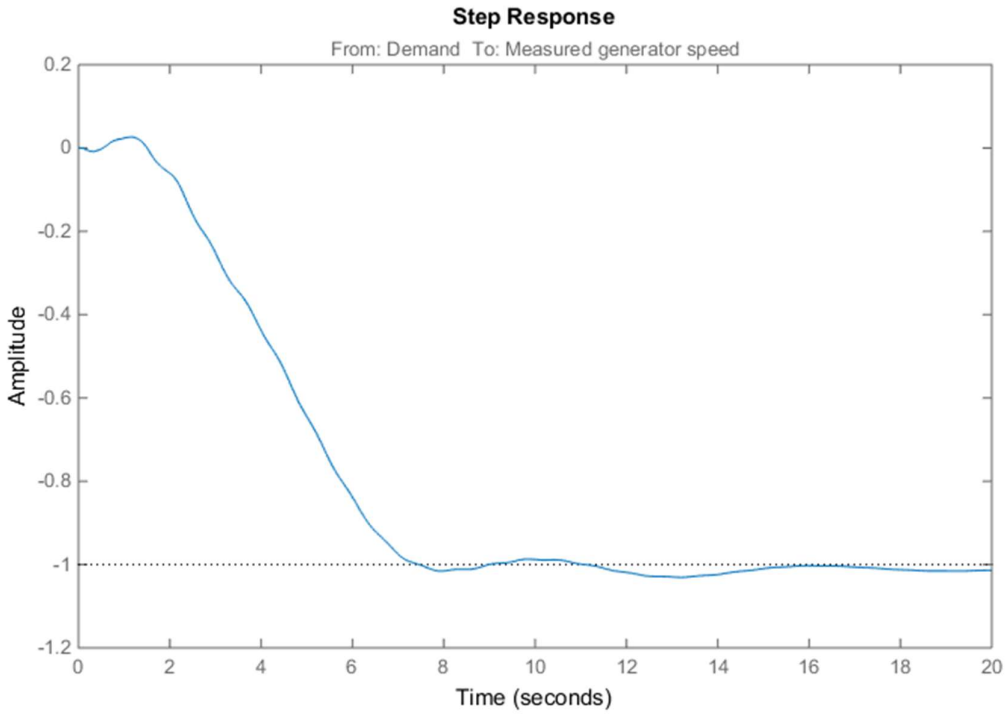


Figure 3-1 step response from generator speed demand to measured generator speed, via the pitch-speed control loop at 10m/s. Loop has been designed to have very little overshoot but still settles within 8 seconds.

This control retune has been implemented by changing the gain schedule values in the blade pitch control section of the DISCON.IN file. This is achieved by editing the contents of the DISCON.IN file via the External Controler “Additional Parameters” field in Bladed. It is important to ensure that PC_GS_n, the number of entries in the gain schedule, matches the number of values in Kp, Ki, Kd and Tf (although Kd and Tf are zeros in this tuning).

Note that additionally, the parameter VS_ControlMode was also set to 2 to give stable behaviour in Bladed.

Limitations

The tower damper (nacelle accelerometer fore-aft direction to pitch demand) loop is disabled in all tests. There is no supervisory control, i.e. start-up, shut-downs, derating, etc.

4 MODEL VERIFICATION

This section describes the steps taken so far to verify the Bladed model.

4.1 Structural frequencies

Blade natural frequencies are compared against the reported values in the IEA report /1/ and the HAWC2 and OpenFAST models presented in the paper /6/. The following should be considered:

- HAWC2 frequencies obtained with Timoshenko with fully populated matrix (H2-FPM)
- OpenFAST frequencies obtained with BeamDyn with 6x6 mass and stiffness matrices

As can be seen in Table 4-1, differences from HAWC2 model are within 7.2% and less than 4% for the first five modes. Discrepancies from the OpenFAST model are slightly larger. It should be noted that the frequencies obtained with HAWC2 and OpenFAST are using a fully populated matrix for mass and stiffness, whereas Bladed is using a modified Timoshenko beam element. The same data is presented visually in Figure 4-1, where similar differences between all three codes are evident.

Table 4-1 Blade frequencies in Hz

Mode	IEA	HAWC2	OpenFAST model	Bladed	Bladed vs.
1st	0.555	0.521	0.517	0.521	0.0% /
1st	0.642	0.619	0.63	0.613	-1.0% / -
2nd		1.559	1.603	1.581	1.4% / -
2nd		1.933	1.964	1.872	-3.2% / -
3rd		3.078	3.506	3.183	3.4% / -
3rd		4.156	4.403	3.863	-7.1% / -
1st		4.475	4.694	4.228	-5.5% / -
4th		5.056	5.307	5.420	7.2% /

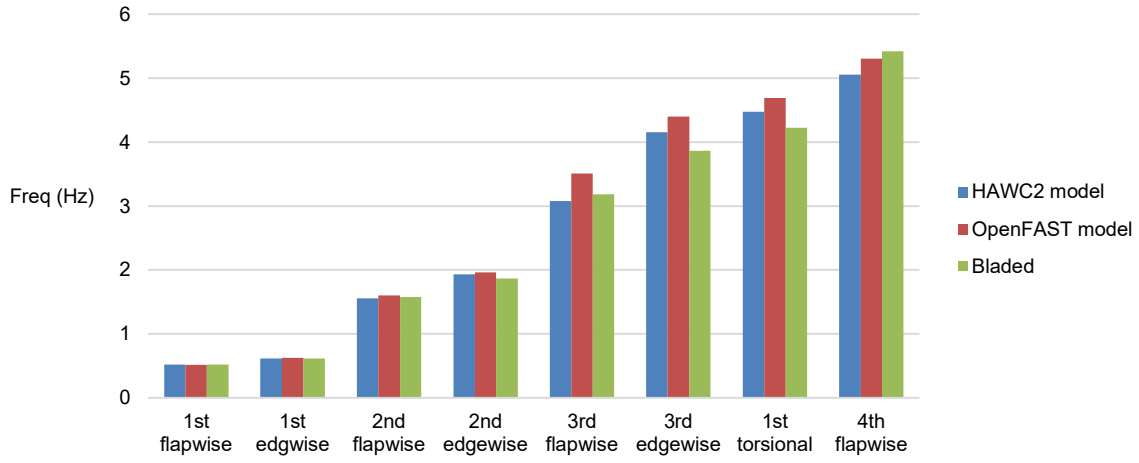


Figure 4-1: Blade frequencies in Hz

A comparison of the first tower frequency against the value in the IEA report is presented in Table 4-2, showing good match.

Table 4-2 Tower frequencies in Hz

Tower mode	IEA definition	Bladed	Difference
1st mode	0.170	0.170	0%

4.2 Steady-state performance

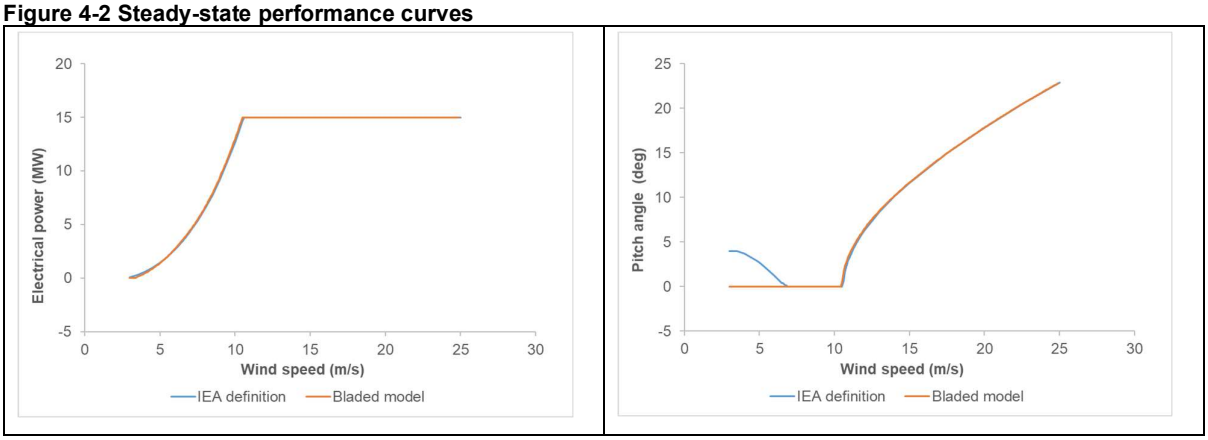
Table 4-3 presents main performance parameters reported in the IEA report and those calculated with the Bladed model with rigid and flexible blades. Steady-state performance curves are provided in Figure 4-2, which have been obtained through a “steady power curve” simulation, that uses the Bladed built-in control functionality with the parameters listed below, instead of the external controller DLL, and where the blade is considered as rigid. The “IEA definition” results are also based on a rigid blade.

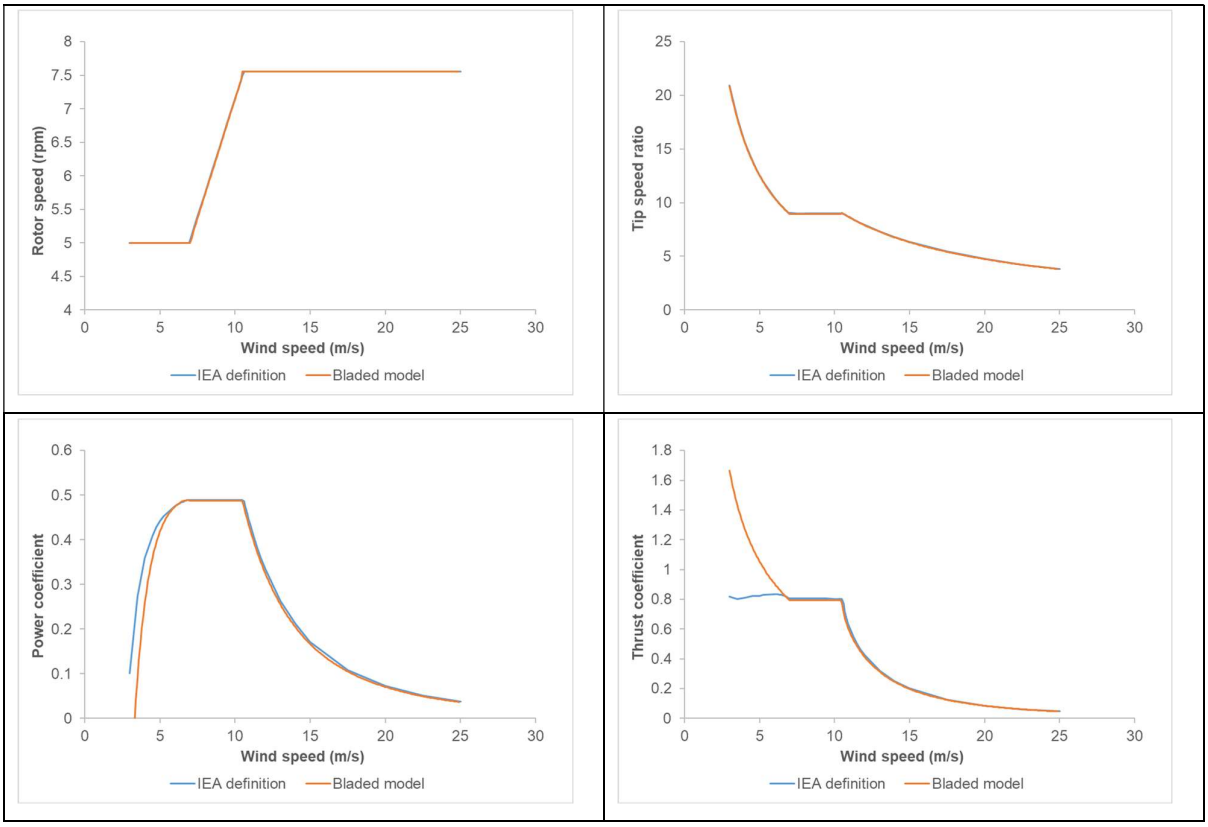
As can be observed, there is very good alignment between the IEA definitions and those calculated with the Bladed rigid model and small differences when blades are flexible. Regarding the steady-state performance curves, the only discrepancy to highlight is the difference in pitch control between cut-in and 7m/s, where Bladed keeps the pitch angle constant while there is small pitching action in the IEA one, which leads to small changes in power and thrust coefficient at those wind speeds. It should be noted that these differences shouldn’t be present when the simulations are run with the project’s external controller DLL.

Table 4-3Turbine specifications.

Parameters	IEA design value /1/	Bladed calculated value (rigid / flexible)	Difference
Power coefficient (C_P)	0.489	0.490 / 0.483	0.3% / -1.2%
Thrust coefficient (C_T)	0.799	0.807 / 0.806	1% / 0.9%
Tip speed ratio (TSR)	9	9.1 / 9.3	1.1% / 3.1%
Blade pitch angle (deg)	0 deg	0 deg / -1.3deg	-

Figure 4-2 Steady-state performance curves





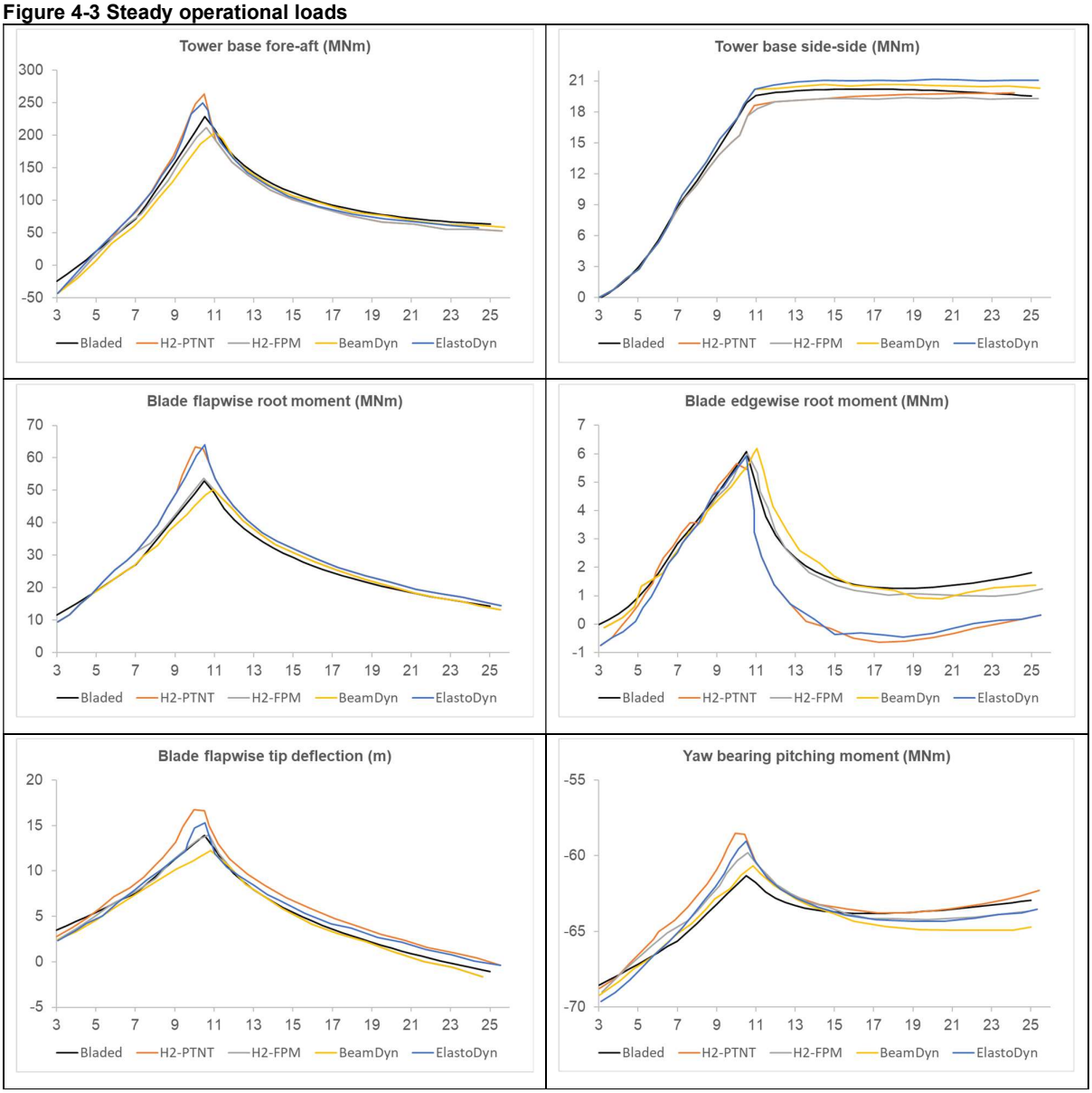
4.3 Steady operational loads

Steady-state operational loads are compared in Figure 4-3 against those calculated with HAWC2 (H2-PTNT, H2-FPM) and OpenFAST (BeamDyn, ElastoDyn) published in the paper /6/. In these aeroelastic codes, the blade is modelled as follows:

- HAWC2 H2-PTNT: Prismatic Timoshenko beam, where torsion is not considered.
- HAWC2 H2-FPM: Prismatic Timoshenko with fully populated 6x6 matrices, allowing for bend-twist coupling.
- OpenFAST ElastoDyn: Euler-Bernoulli beam theory with assumed modes, where torsion is not considered.
- OpenFAST BeamDyn: Beam theory with Legendre spectral finite elements, defined with fully populated 6x6 matrices

As can be observed, Bladed loads follow very closely the HAWC2 H2-FPM and FAST BeamDyn results. This is is the expected result as these 3 models have a non-linear blade deflection model, and include torsional flexibility.

Figure 4-3 Steady operational loads

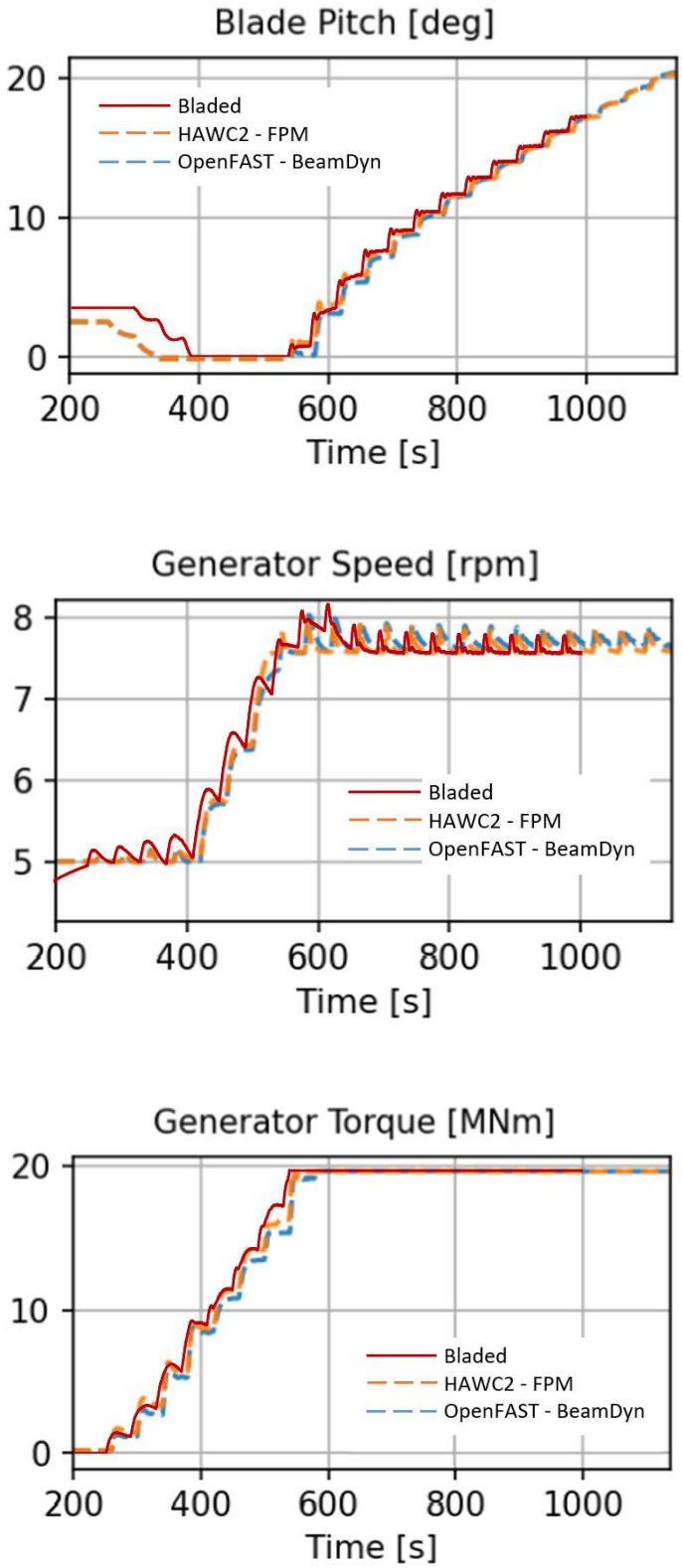


4.4 Power production wind steps

The closed-loop dynamic response of the Bladed model was tested using the ROSCO controller. Simulations were run with wind speed steps ranging from from 3 to 22m/s (in steps of 1m/s and 40s duration for each step), with the aim to reproduce the results reported in the paper /6/

Figure 4-4 presents pitch angle, generator speed and generator torque responses for the new version of ROSCO (2.3.0) controller overlayed with HAWC2/OpenFAST curves from the paper /6/. Generally a good agreement is seen between Bladed and the other codes.

Figure 4-4 Pitch angle, generator speed and generator torque step-wind response, Bladed results overlayed with graphs from paper /6/.



4.5 Loads verification

Extreme sub-set of loads are available in the IEA report /1/. The load cases included in the sub-set are power production (DLC1.1, 1.3, 1.4 and 1.5) and idling (DLC6.1 and DLC6.3).

Table 4-4, Figure 4-5 and Figure 4-6 show comparisons between the IEA load and deflection values and those calculated with Bladed. No safety factors have been considered.

Generally a good agreement is observed. It has not been possible to investigate the source of the differences as the original simulation time histories from OpenFAST were not available.

Table 4-4 Comparison of load components

Parameters	IEA report value /1/	Bladed calculated value	Difference
Blade root flapwise (MNm)	89 (DLC1.5)	76 (DLC1.5)	-13.7%
Blade root edgewise (MNm)	30 (DLC1.4)	36 (DLC6.3)	21.2%
Tower base side-side (MNm)	416 (DLC6.3)	407 (DLC6.1)	-2.1%
Tower Base fore-aft (MNm)	400 (DLC1.4)	430 (DLC1.4)	7.6%
Blade tip flap deflection (m)	22.8 (DLC1.4)	19.7 (DLC1.5)	-13.8%
Blade tip edge deflection (m)	7.6 (DLC6.1)	6.1 (DLC6.1)	-19.3%
Tower top side-side deflection (m)	1.6 (DLC6.1)	1.95 (DLC6.1)	25.0%
Tower top fore-aft deflection (m)	0.9 (DLC1.4)	1.90 (DLC1.4)	103.8%

Figure 4-5 Comparison of blade root and tower base moments

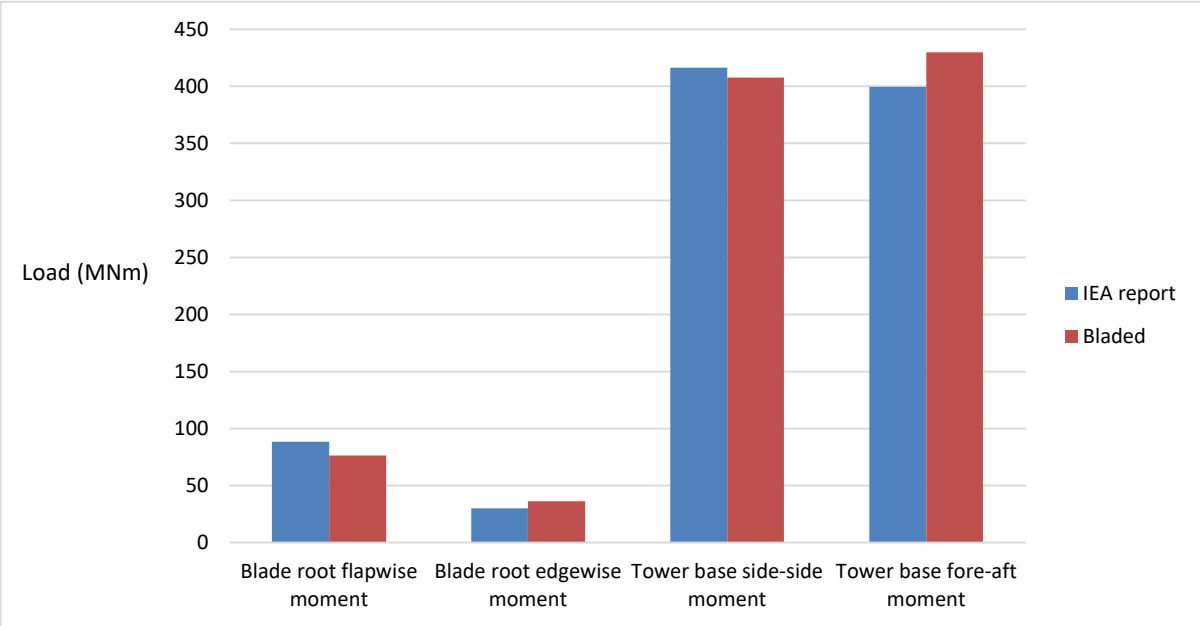
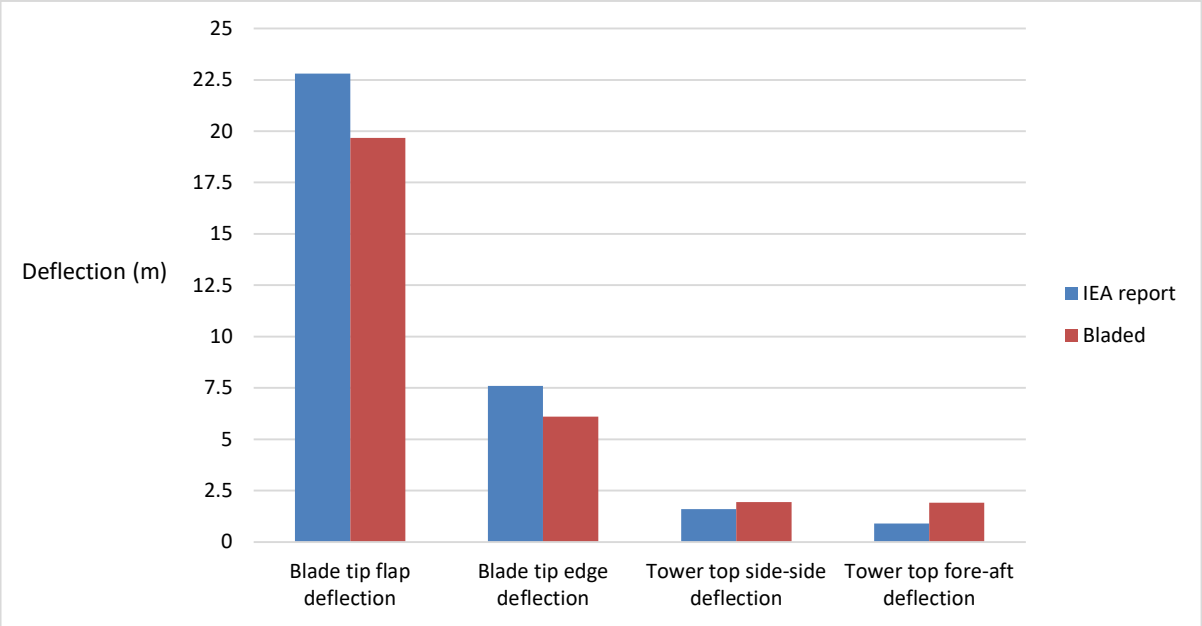


Figure 4-6 Comparison of blade tip and tower top deflections



5 MODEL USAGE

The delivery package for this model contains the following components

1) The turbine model Bladed project file: *15MW_NREL_DNV_v1.PRJ*

2) Some dependencies that the PRJ file refers to:

    a) The ROSCO controller version 2.3.0 compiled in 32 bit: *libdiscon_2.3.0.dll.*

    b) A file describing inputs to the ROSCO controller wind speed estimator component: *Cp_Ct_Cq.IEA15MW*.

    <u>Note that the references to these dependencies must be updated in the PRJ file before the model can be run on your local machine.</u>

3) The *DISCON.in* file that describes the ROSCO controller parameters for this turbine, retuned for use with Bladed. Note that this information is also contained in the PRJ file.

4) Technical note "Running Bladed simulations with ROSCO controller". This is for information only as the PRJ file and ROSCO controller are already set up in the other provided files.

5) This report

Items 2a and 2b are simply copies of files provided on the IEA Task 37 GitHub site, provided in the delivery package for users' convenience.

6 REFERENCES

/1/ NREL/TP-5000-75698, Definition of the IEA Wind 15-Megawatt Offshore Reference Wind Turbine, March 2020.

/2/ IEA Wind Task 37/IEA-15-240-RWT GitHub site, GitHub - IEAWindTask37/IEA-15-240-RWT: 15MW reference wind turbine repository developed in conjunction with IEA Wind, last accessed on 24th June 2021.

/3/ Documentation folder, IEA-15-240-RWT/Documentation at master · IEAWindTask37/IEA-15-240-RWT · GitHub

/4/ HAWC2 model data, IEA-15-240-RWT/HAWC2 at master · IEAWindTask37/IEA-15-240-RWT · GitHub, last accessed on 24th June 2021.

/5/ OpenFAST model data, IEA-15-240-RWT/OpenFAST at master · IEAWindTask37/IEA-15-240-RWT · GitHub, last accessed on 24th June 2021.

/6/ Jennifer Rinker et al, Comparison of loads from HAWC2 and OpenFAST for the IEA Wind 15 MW Reference Wind Turbine, The Science of Making Torque from Wind, 2020.

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